PART | ONE

SYSTEM PHILOSOPHY

The Name of the Devil

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Chapter | one

How the Game Is Evolving

The most stubborn habits, which resist change with the greatest tenacity, are those that worked well for a space of time and led to the practitioner being rewarded for those behaviors. If you suddenly tell such persons that their recipe for success is no longer viable, their personal experience belies your diagnosis. The road to convincing them is hard. It is the stuff of classic tragedy. [1]

The Dow Jones Industrial Average recently marked its 100th anniversary. Of the original companies listed in 1896 only GE had survived to join in the celebration. In the mid-1960s, Jean-Jacques Shreiber, in his best-selling book, *American Challenge* (1967), told his fellow Europeans: "Swallow your pride, imitate America, or accept her dominance forever." But in late 1970s, it was "Japan Inc." that somehow posed the greatest competitive challenge to corporate America. It took 300% devaluation of the dollar to ward off this challenge.

Fourteen of the 47 companies exemplified in Tom Peters' much-acclaimed book of the 1980s, *In Search of Excellence* (1982), lost their luster in less than four years, at least in the sense that they had suffered serious profit erosion.

The collapse of savings and loans and real estate, along with the fall of the defense industry in the late 1980s, could have led to a disastrous 1990s, but counterintuitively, these phenomena resulted in a restructuring of the financial and intellectual resources in America, which may very well have been a coproducer of one of the longest periods of economic expansion and prosperity in America. Ironically, in mid-1998, worries about Japan's economy were the nagging concerns of American investors. Collapse of the dotcom bonanza (late 1999 and early 2000) and the housing bubble and the subprime and financial systems fiasco led to the troubling question: What is going on?

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[1]Charles Hampden-Turner and Linda Arc, *The Raveled Knot: An Examination of the Time-to-Market Issue at Analog's Semi-conductor Division*, unpublished internal report.

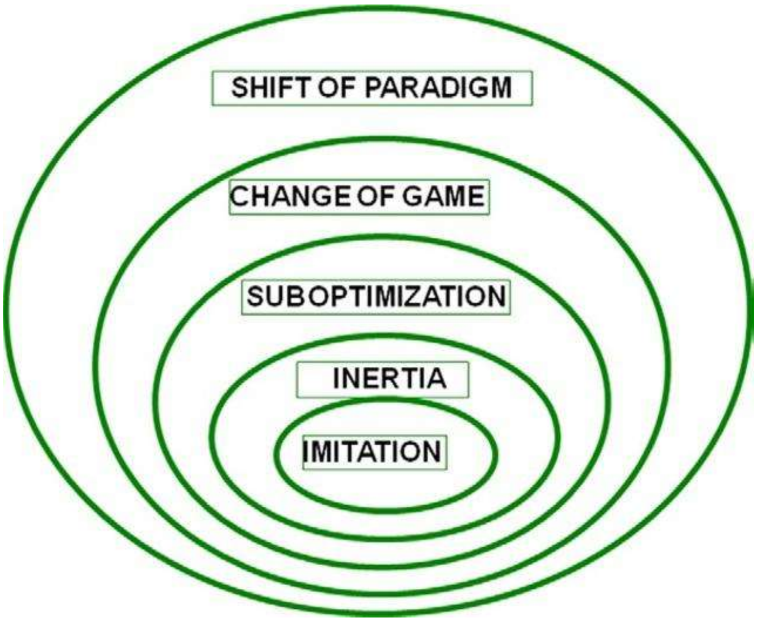

FIGURE 1.1 Hierarchy of forces that erode competitive advantage.

The game keeps changing, but this is hardly news. By now it is a well-known and even a tired secret that what contributes to the fall of so many great enterprises is that somehow their recipe for success becomes ineffective. There seems to be a devil at work here, and the name of this devil is success.

Each one of us can recall cases of great powers, nations, organizations, or personalities rising and falling. This phenomenon occurs all too frequently to be dismissed as coincidental. So what underlying forces convert success to failure? Let us start with the following observation. The forces that make a failure out of success form a five level hierarchy (see Figure 1.1). Each level represents a distinct tendency, but together they form an interactive whole in which higher levels provide the context for the lower levels. At each level success plays a critical but different role.

1.1 IMITATION

Operating at the first level, imitation is the most basic force. Competitive advantage is by definition a distinction. Successful distinctions, in time, are eroded by imitation. At that point, exceptions become norms and lose their advantage.

Although imitation has been present at all times, today its significance for American business has changed by an order of magnitude. Advances in information technology, communication, and reverse engineering have increased the product technology's vulnerability to imitation. Any technological distinction in a given product is now fair game for potential imitators who can learn, copy, and reproduce it in practically no time. Such easy imitation has been significant for American industry. While product technology has traditionally been the cornerstone of the American competitive game, countries with an advantage in process technology have gained a dual advantage.

First, it is difficult to copy a distinction in process technology because its critical elements are knowledge workers. Second, competency in a process technology makes it simpler to transfer knowledge from one context to another, easing the operationalization of new knowledge. The results are dramatic: much faster time-to-market performance, a lower break-even point, better product variety, and faster response to change.

In the late 1970s, a well-known equipment company in America realized it had a 40% cost disadvantage in comparison with its direct Japanese competitor. The company, ironically, was the technological leader in the lift truck industry. Its cost structure was 40% raw material, 15% direct labor, and 45% overhead. Overhead (transformation cost) was simply calculated as 300% of direct labor.

The company decided to reduce the cost by 20%. It was assumed that a 5% reduction in direct labor would automatically reduce overhead by another 15%, resulting in a 20% cost reduction. After a whole year of struggle, direct labor was reduced to 10% without any reduction in the overhead. When we were asked to deal with the situation, this was our first reaction: Why does anyone want to reduce the cost by 20% when there is a 40% cost disadvantage? Where did the 40% cost advantage come from? It was obvious that even if the workers gave up all of their wages the company would not survive.

Then we realized that the competitive product only used 1,800 parts while our product employed 2,800. The difference in the number of the parts perfectly explained the difference in cost. The surprising element in all of this was that a lower number of parts was achieved by the competition by utilizing technologies that were developed by our client over the last 10 years. The problem was that our client had patched each one of its newly developed technologies into an old platform, which resulted in a complex and inefficient product, whereas the competition started from a clean slate and took full advantage of the potentials that each technology offered.

The moral of this story is that once in a while one should pause and reflect on oneself and begin anew.

1.2 INERTIA

Inertia is responsible for all of the second level tendencies and behaviors that delay reactions to technological breakthroughs. For example, sheer inertia by the Continental Can Company provided the opportunity for two-piece can technology to replace the three-piece can technology and destroy the once mighty Continental Can. Five hundred factories all over the United States and 45% share of the three-piece can market could not prevent a delayed reaction to two-piece technology from destroying Continental Can in fewer than three years.

Ironically, the likelihood that an organization will fail to respond to a critical technological break is directly proportional to the level of success it had achieved in a previously dominant technology. In other words, the more success an organization has with a particular technology, the higher its resistance to the prospect of change. The initial reaction is always denial. We do have an amazing capacity for denial in the face of undeniable events, but the real danger arises when the organization finally decides to patch things up. Patching wastes critical time. It provides the competition with a window of opportunity to disseminate the new technology and dominate the market. Patching, moreover, increases the cost of the operation and reduces the quality of the output, producing a double jeopardy.

1.3 SUBOPTIMIZATION

Exaggeration — the fallacy that if "X" is good more "X" is even better — is at the core of the third level processes that effectively destroy a proven competitive advantage. A tendency to push one's strength to its limits transforms the strength into a destructive weakness. Unfortunately, many stories follow the same line: a winning formula gains adulation, and the heroes or heroines who shaped it become the sole authorities. One right answer prevails. An increasingly monolithic culture produces an ever-decreasing set of alternatives and a narrow path to victory. This limited set redefines the corporate culture, the assumptions, the premises, and the common wisdom that bounds or frames a company's understanding of itself and its industry and drive its competitive strategy.

An interesting treatment of this phenomenon can be found in Danny Miller's book, *The Icarus Paradox* (1990). Miller refers to Icarus of Greek mythology who became emboldened to fly higher and higher until he came so close to the sun that his wax wings melted and he plunged to his death. Miller explains how craftsmanship and productive attention to detail by the Digital Equipment Corporation turned into an obsession with minutia and technical tinkering. Exaggeration was also at work when the innovative capability of CDC and Polaroid escalated into high-tech escapism and technical utopia. Miller's list of firms that have been trapped by this phenomenon includes IBM, Texas Instruments, Apple Computer, General Motors, Sears, and many of the most acclaimed American corporations.

1.4 CHANGE OF THE GAME

Change of the game, or transformation of the problem, is at the heart of a counterintuitive process that converts success into failure. In other words, the act of playing a game successfully changes the game itself. Failure to appreciate the consequences of one's success and tenacity in playing the good old game are what create tragedies. Once success is achieved, or a problem is effectively dissolved, the concerns associated with that problem are irreversibly affected. Dissolving a problem transforms it and generates a whole new set of concerns. That is why the basis for competition changes and a new competitive game emerges as soon as a competitive challenge is met.

The role of success is quite different in the third and fourth level processes. When it is exaggerated (third level), success works against the nature of the solution and diminishes its effectiveness. By contrast, success in handling a challenge (fourth level) transforms the nature of the problem. In other words, it changes the game. Henry Ford's success in creating a mass production machine effectively dissolved the production problem. A familiar concern for production was replaced with an unfamiliar concern for markets. The once unique ability to mass-produce lost its advantage through widespread imitation. This event changed the competitive game from concern for production to concern for markets, which required an ability to manage diversity and growth.

Henry Ford's refusal to appreciate the implication of his own success and his unwillingness to play the new game ("they can have any color as long as it is black") gave Alfred Sloan of GM the opportunity to dominate the automotive industry. Sloan's concept of product-based divisional structure turned out to be an effective design for managing growth and diversity. The new game, artfully learned and played by corporate America, became the benchmark for the rest of the world to copy (Womack, 1990).

In an attempt to duplicate the American system, Ohno, the chief engineer of Toyota, came up with yet another new design. His introduction of the lean production system changed the performance measures by more than an order of magnitude. While it took the American auto industry three days to change a die, Toyota could do it in only three minutes. Once again, success transformed the game. This time the differentiating factors were flexibility and control.

But corporate America was too overwhelmed and overjoyed by its own success to even notice the emergence of the new game. This inattentiveness provided Japan with an opportunity to launch a slow but effective challenge. The insidious manner in which the new game evolved underscores another important principle of systems dynamics, which is exemplified by the story of the frog that boiled to death by sitting happily in water that gradually grew hotter.

Examples of the change of the game can also be found in politics. Although the success of the Persian Gulf War boosted the approval rating of President Bush to an unprecedented level, it inadvertently cost him the election. The triumph of his foreign policy caused the nation to shift its concern from national security to domestic economy. Failure to understand the implication of this change converted the success to failure.

Recognizing that success changes the game, think what the phenomenal success of information technology means. Success marks the beginning of the end of the Information Era. Competitive advantage is increasingly shifting away from having access to information to generating knowledge and, finally, toward gaining understanding.

1.5 SHIFT OF PARADIGM

The cumulative effects of imitation, inertia, suboptimization, and change of the game ultimately manifest themselves in the fifth force — a shift of paradigm.

A shift of paradigm can happen purposefully by an active process of learning and unlearning. It is more common that it is a reaction to frustration produced by a march of events that nullify conventional wisdom. Faced with a series of contradictions that can no longer be ignored or denied and/or an increasing number of dilemmas for which prevailing mental models can no longer provide convincing explanations, most people accept that the prevailing paradigm has ceased to be valid and that it has exhausted its potential capacity.

This is a twilight zone where Stafford Beer's (1975) aphorism rings true: "Acceptable ideas are competent no more and competent ideas are not yet acceptable." It is where powerful threats and opportunities emerge; where the great organizations rise and fall.

Eventually, it takes the exceptional courage of a few to question the conventional wisdom and point to the first crack in it. Thus begins a painful struggle whose end result is reconceptualization of critical variables into a new ensemble with a new logic of its own.

Shifts of paradigm can happen in two categories: a change in the nature of reality or a change in the method of inquiry. Also possible, however, is a dual shift involving both dimensions. The significance and impact of any paradigm shift cannot be overestimated, but facing a dual shift is an even more formidable challenge. It tests the outer limits of human capacity to comprehend, communicate, and confront the problematic. For example, the shift of paradigm from a mechanical to a biological model, despite its huge impact, represented a unidimensional shift in our understanding of the nature of organization. It happened in the context of analytical inquiry (Figure 1.2).

We are now facing the challenge of a dual shift. Not only has there been a shift of paradigm in our understanding of the nature of the beast — from our conception of an organization as a biological model to a sociocultural model — but there has also been a profound shift in our assumption regarding the method of inquiry, the means of knowing, from *analytical thinking* (the science of dealing with *independent* sets of variables) to *holistic thinking* (the art and science of handling *interdependent* sets of variables). The complementary nature of these two dimensions is at the core of both understanding how the game is evolving and identifying the drivers for change.

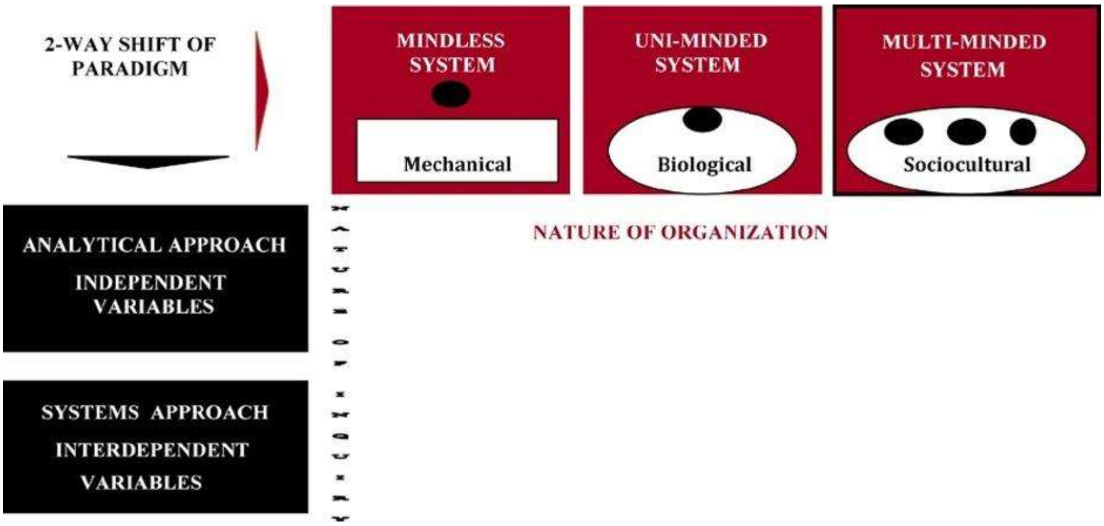

FIGURE 1.2 Shifts of paradigm.

1.6 INTERDEPENDENCY AND CHOICE

While the organization as a whole is becoming more and more *interdependent*, the parts increasingly display choice and behave *independently*. The resolution of this dilemma requires a dual shift of paradigm.

The first shift results in the ability to see the organization as a multi-minded, sociocultural system, a voluntary association of purposeful members who have come together to serve themselves by serving a need in the environment.

The second shift helps us see through chaos and complexity and learn how to deal with an interdependent set of variables. Failure to appreciate the significance of this dual change results in excessive structural conflict, anxiety, a feeling of impotency, and resistance to change. Unfortunately, prevailing organizational structures, despite all the rhetoric to the contrary, are designed to prevent change. Dominant cultures by default keep reproducing the same non-solutions all over again. This is why the experience with corporate transformation is so fraught with frustration. The implicitness of the organizing assumptions, residing at the core of the organization's collective memory, is overpowering. Accepted on faith, these assumptions are transformed into unquestioned practices that may obstruct the future. Unless the content and implications of these implicit, cultural codes are made explicit and dismantled, the nature of the beast will outlive the temporary effects of interventions, no matter how well intended.

1.6.1 On the Nature of Organization: The First Paradigm Shift

To think about any thing requires an image or a concept of it. To think about a thing as complex as an organization requires models of something similar, something simpler, and something more familiar. The three models represent the successive shift in our understanding of the nature of the organization, from a mindless mechanical tool, to a uni-minded biological being and, finally, to a multi-minded organized complexity.

1.6.1.1 MINDLESS SYSTEM — A MECHANISTIC VIEW

The mechanistic view of the world that evolved in France after the Renaissance maintains that the universe is a machine that works with a regularity dictated by its internal structure and the causal laws of nature. This worldview provided the basis not only for the Industrial Revolution but also for the development of the machine mode of organization (Gharajedaghi and Ackoff, 1984).

In the early stages of industrialization, machines replaced agricultural workers by the thousands. The reservoir of an unemployable army of unskilled agricultural workers threatened the fabric of Western societies. Then came a miracle, the ingenious notion of organizations. It was argued that in the same way a complicated tractor is built by parts, each performing only a simple task of horizontal, vertical, and circular motions, an organization could be created in such a manner that each person performs only a simple task. The mechanistic mode of organization was born as a logical extension of this conception and became instrumental in converting the army of unskilled agricultural laborers to semi-skilled industrial workers (Figure 1.3).

The impact of this simple notion of organizations was so great that in one generation it created a capacity for the production of goods and services that surpassed the cumulative capacity of mankind. The essence of

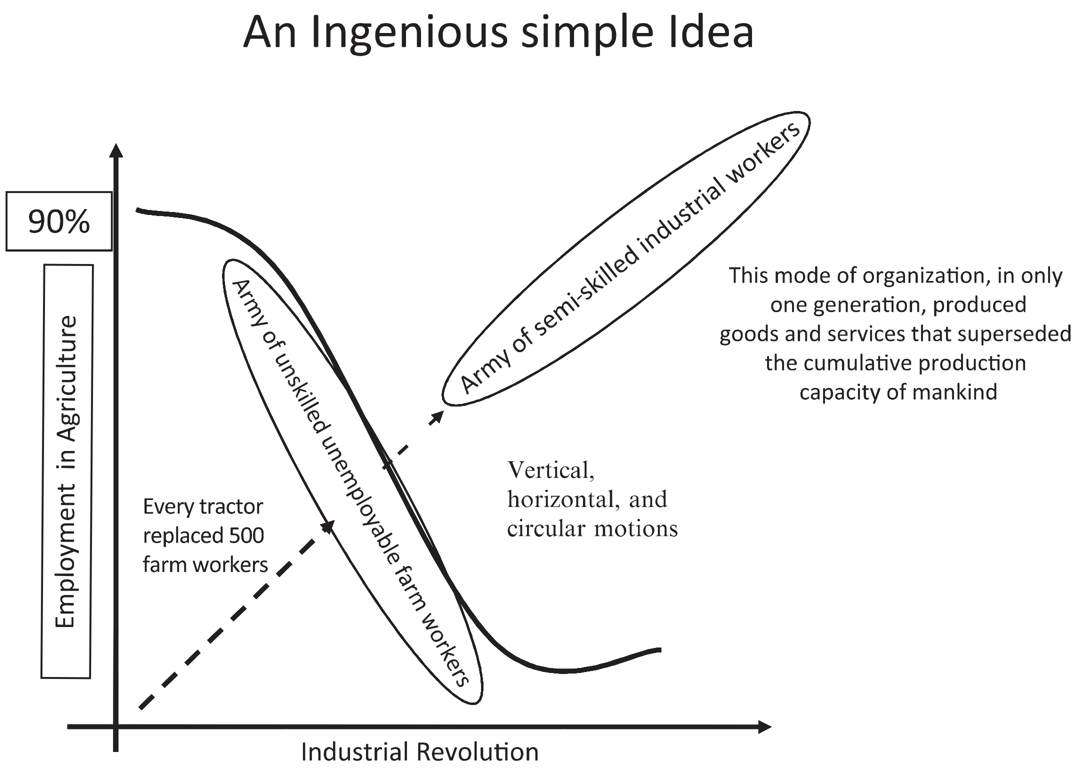

FIGURE 1.3 Machine mode of organization.

the machine mode of organization is simple and elegant. An organization is a mindless system; it has no purpose of its own. It is a tool with a function defined by the user, an instrument for the owner to use to achieve his goal of making profit. The important attribute of this tool is its reliability, and its performance criterion is simply efficiency. The principle that parts should not deviate is at the core of the glamour of tidiness, efficiency, controllability, and predictability of its operation. The parts of a mindless mechanical system, just like the whole, have no choice. Its structure is designed into it, leaving it with no ability to restructure itself. The system functions reactively and can operate effectively only if its environment remains stable or has little effect on it.

1.6.1.2 UNI-MINDED SYSTEMS — A BIOLOGICAL VIEW

The biological thinking or living systems paradigm, which led to the concept of the organization as a uni-minded system, emerged mainly in Germany and Britain, but then caught fire in the United States. The underlying assumptions and principles of the biological mode of organizations are also simple and elegant: an organization is considered a uni-minded living system, just like a human being, with a purpose of its own. This purpose, in view of the inherent vulnerability and unstable structure of open systems, is survival. To survive, according to conventional wisdom, biological beings have to grow. To do so they should exploit their environment to achieve a positive metabolism.

In organizational language, this means that growth is the measure of success, the single most important performance criterion, and that profit is the means to achieve it. Therefore, in contrast to the machine mode, in which profit is an end in itself, profit, for the biological mode, is only a means to an end. The association of profit with growth, considered a social good, gives profit the much needed social acceptability and status compatible with the American way of life.

Although uni-minded systems have a choice, their parts do not. They operate based on cybernetics principles as a homeostatic system, reacting to information in the same way as a thermostat. As a matter of fact, the beauty of a uni-minded system is that the parts do not have a choice and react only in a predefined manner to the events in their environment.

For example, my heart cannot decide on its own that it does not want to work for me. My stomach will not get suspicious, thinking "the liver is out to get me." No consciousness, no choice, no conflict. The operation of a uni-minded system is totally under the control of a single brain, the executive function, which, by means of a communication network, receives information from a variety of sensing parts and issues directions that activate relevant parts of the system. It is assumed that a malfunctioning of any normal uni-minded system is due to a lack of information or noise in the communication channel. Therefore, the perceived answer for most of the problems is more information and better communication. However, if parts of a system develop consciousness and display choice, the system will be in real trouble. Imagine for a moment that the thermostat in your room suddenly develops a mind of its own — when it receives information about the temperature in the room it decides it does not like it and wants to sleep on it. The undeniable result is a chaotic air conditioning system.

When parts display choice, the central issues become conflict and the ability to deal with it. However, as long as paternalism is the dominant culture, the imperatives of "father knows best" or "give the apple to your sister" become an effective way to handle conflict. Paternalism best approximates the essential characteristics of a uni-minded system, and it creates powerful organizations. Corporate giants such as Ford, DuPont, General Motors, and IBM owe much to their paternalistic founding fathers.

1.6.1.3 MULTI-MINDED SYSTEM — A SOCIOCULTURAL VIEW

Multi-minded systems are exemplified by social organizations. A sociocultural view considers the organization a voluntary association of purposeful members who manifest a choice of both ends and means. This is a whole new ball game. Behavior of a system whose parts display a choice cannot be explained by mechanical or biological models. A social system has to be understood on its own terms.

The critical variable here is *purpose*. According to Ackoff (1972), an entity is purposeful if it can produce (1) the same outcome in different ways in the same environment and (2) different outcomes in the same or a different environment. Although the ability to make a choice is necessary for purposefulness, it is not sufficient. An entity that can behave differently but produce only one outcome in all environments is goal-seeking, not purposeful. Servo-mechanisms are goal-seeking, but people are purposeful. As a purposeful system, an organization is part of a larger purposeful whole — the society. At the same time, it has purposeful individuals as its own members. The result is a hierarchy of purposeful systems of three distinct levels. These three levels are so interconnected that an optimal solution cannot be found at one level independent of the other two. Aligning the interest of the purposeful parts with each other and that of the whole is the main challenge of the system.

In contrast to machines, in which integrating of the parts into a cohesive whole is a one-time proposition, for social organizations the problem of integration is a constant struggle and a continuous process. Effective integration of multilevel purposeful systems requires that the fulfillment of a purposeful part's desires depends on fulfillment of the larger system's requirements, and vice versa. In this context, the purpose of an organization is to serve the purposes of its members while also serving the purposes of its environment.

The elements of mechanical systems are *energy-bonded*, but those of sociocultural systems are *information-bonded*. In energy-bonded systems, laws of classical physics govern the relationships among the elements.

Passive and predictable functioning of parts is a must, until a part breaks down. An automobile yields to its driver regardless of his expertise and dexterity. If a driver decides to run a car into a solid wall, the car will hit the wall without objection. Riding a horse, however, presents a different perspective. It matters to the horse who the rider is, and a proper ride can be achieved only after a series of information exchanges between the horse and the rider. Horse and rider form an information-bonded system in which guidance and control are achieved by a second degree agreement (agreement based on a common perception) preceded by a psychological contract.

The members of a sociocultural organization are held together by one or more common objectives and collectively acceptable ways of pursuing them. The members share values that are embedded in their culture. The culture is the cement that integrates the parts into a cohesive whole. Nevertheless, since the parts have a lot to say about the organization of the whole, consensus is essential to the alignment of a multi-minded system.

1.7 ON THE NATURE OF INQUIRY

1.7.1 The Second Paradigm Shift

Classical science is preoccupied with *independent variables.* It assumes that the whole is nothing but the sum of the parts. Accordingly, to understand the behavior of a system we need only to address the impact that each independent variable has on that system (Figure 1.4).

Handling independent variables is the essence of analytical thinking, which has remained intact in all three contexts: physical, biological, and social. To share in the glory of classical science, both biological and social sciences opted to use the analytical method with no deviation. This might help explain why a whole set of phenomena, known as type II (emergent) property, has been conveniently ignored. Properties like love, success, and happiness do not yield to analytical treatment.

However, increasingly we are finding out that our independent variables are no longer independent and that the neat and simple construct that served us so beautifully in the past is no longer effective. The following experience illustrates this point.

Ford Motor Company was one of the first American corporations to embark on the quality movement. "Quality is job one" was the theme, and the operating units were encouraged to use continuous improvement to

$$Y = F\,(x1,\ x2,\ x3,\ x4,\ \ldots)$$

FIGURE 1.4 Independent variables.

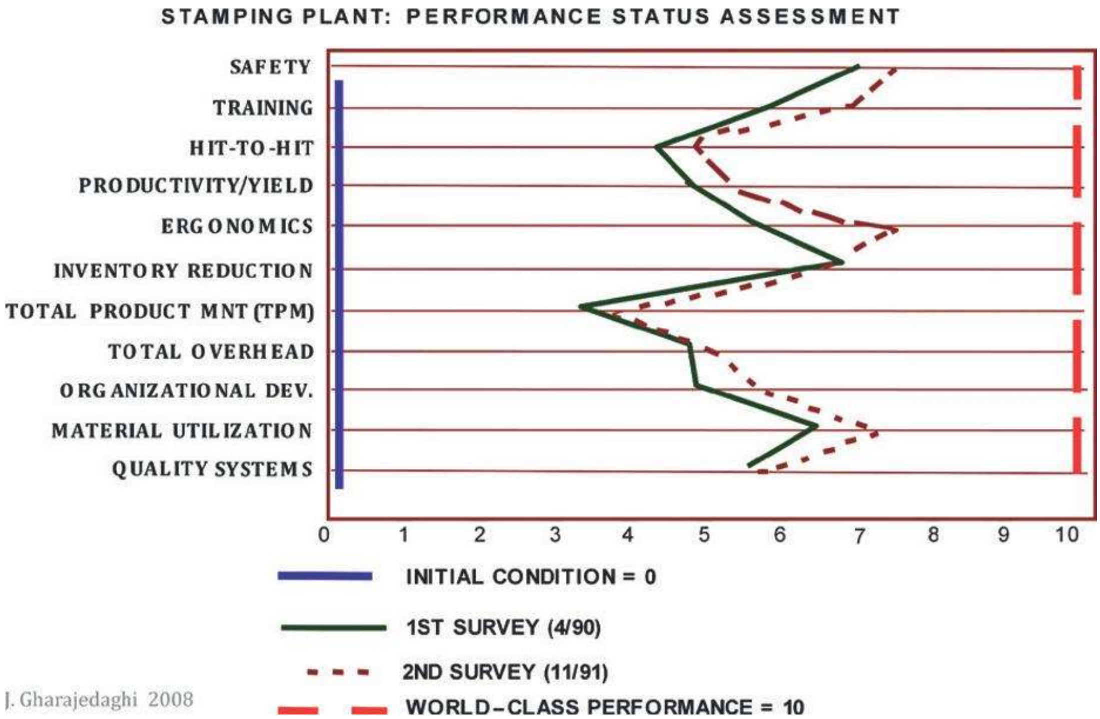

FIGURE 1.5 Woodhaven stamping plant's quality variables.

achieve world-class performance. Following the lead was Ford's Woodhaven stamping operations, which identified eleven areas of improvement (Figure 1.5).

Initial (or baseline) measures in each area were designated as 0 and world-class performance as 10. The company established a detailed and comprehensive program to go from 0 to 10 in three years. Initially, significant improvement was recorded, but the operation reached a plateau after only 18 months.

Even doubling the efforts to improve the selected variables' performance failed to produce any further change. After 36 months of intense effort, the operation remained at the midway point of its goals, well short of the benchmark, world-class performance (Figure 1.6).

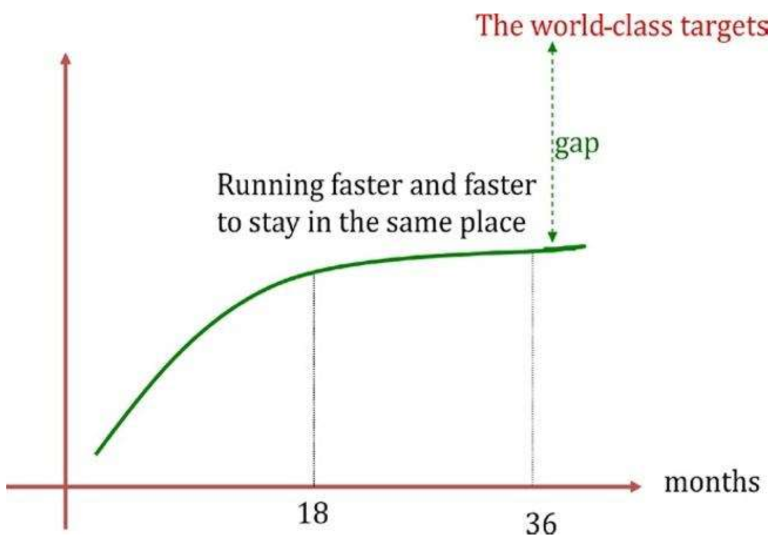

FIGURE 1.6 Reaching a plateau before getting to the target.

At the time I was teaching in the Ford Executive Development Program. Mr. Vic Leo, Program Director, introduced me to Mr. E.C. Galinis, Plant Manager of the Woodhaven operation, who shared his frustration with me. After spending a few days in the plant, I concluded that the Woodhaven operation had used up all of its slack and was now faced with a set of interdependent variables that could be improved only with a redesign of the total operation (Figure 1.7).

As Figure 1.7 demonstrates, a given design may contain some slack between variables. This permits us to deal with each variable separately as though it were an independent variable. The performance of each variable can be improved independently until the slack among them is used up. Then the perceived set of independent variables changes to a formidable set of interdependent variables. Improvement in one variable would come only at the expense of the others.

Using the conventional approach to deal with this type of situation would be like riding a treadmill. One needs to keep running faster and faster to stay in the same place. In Ford's case, the existing design of Woodhaven operations had reached its highest potential, unfortunately far below the world-class performance. To reach the performance goals, the operation would have to be redesigned, and this was done. A new design helped the operation not only to reach the target goal, but also to surpass it by a wide margin in six months.

An independent set of variables is, therefore, a special case of a more general scheme of *interdependency*. As systems become more and more sophisticated, the reality of interdependency becomes more and more pronounced (see Figure 1.8).

Understanding interdependency requires a way of thinking different from analysis. It requires systems thinking. And analytical thinking and systems thinking are quite distinct.

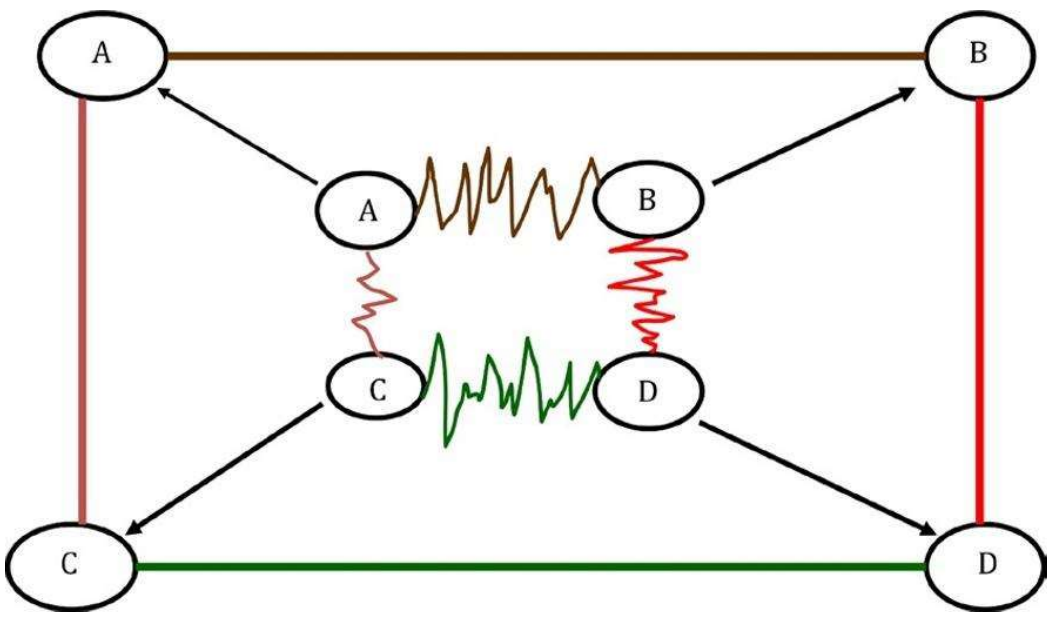

FIGURE 1.7 Using up the slack among interdependent variables.

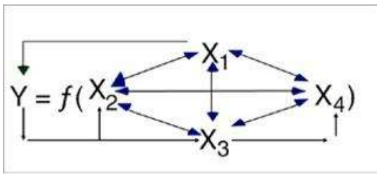

FIGURE 1.8 Interdependent variables.

Analysis is a three-step thought process. First, it takes apart that which it seeks to understand. Then it attempts to explain the behavior of the parts taken separately. Finally, it tries to aggregate understanding of the parts into an explanation of the whole. Systems thinking uses a different process. It puts the system in the context of the larger environment of which it is a part and studies the role it plays in the larger whole.

Analytical approach has remained essentially intact for nearly four hundred years, but systems thinking has already gone through three distinct generations of change:

- The first generation of systems thinking (operations research) dealt with the challenge of *interdependency* in the context of mechanical (deterministic) systems.
- The second generation of systems thinking (cybernetics and open systems) dealt with the dual challenge of *interdependency* and *self-organization* (neg-entropy) in the context of living systems.
- The third generation of systems thinking (design) responds to the triple challenge of *interdependency, self-organization,* and *choice* in the context of sociocultural systems.

In addition to being purposeful, social organizations are living systems; therefore, like all living systems, they are neg-entropic and capable of self-organization. They create order out of chaos. Biological systems primarily self-organize through genetic codes, and social systems self-organize through cultural codes. The DNA of social systems is their culture.

Social systems, however, can be organized either by default or by design. In default, the beliefs, assumptions, and expectations that underlie the system go unexamined. In design, the beliefs, assumptions, and expectations are made explicit, being constantly examined and monitored. The third generation of systems thinking therefore has to deal not only with the challenge of interdependency and choice, but also with the implications of cultural prints reproducing the mess, or the existing order, all over again by default. This is why design, along with participation, iteration, and second-order learning, is at the core of the emerging concept of systems methodology.

Details of this exciting concept are explored in Part Three of this book, which develops an operational definition of systems thinking. The remainder of this chapter explores implications of the dual paradigm shift in the context of six distinct competitive games.

1.8 THE COMPETITIVE GAMES

Each of the competitive games discussed in this section corresponds to a given paradigm in the following matrix (Figure 1.9). Together, these games have dominated the management scene for the better part of the past century. Each has produced an order-of-magnitude change in performance measures, and each has had a profound effect on our lives.

Each paradigm has its own unique mode of organization, and every mode of organization, by virtue of its requirement for specific talents, creates its own clique and privileged members. These members often translate their privileges into power and influence. The higher the level of success, the greater the stake in continuing an existing order and the higher the resistance to change. Unfortunately, the inability to change an outdated mode of organization is as tragic for the viability of a corporation as the consequence of missing a technological break is for the viability of a product line.

1.8.1 Mass Production — Interchangeability of Parts and Labor

Mass production resulted directly from the machine mode of organization. Henry Ford's success in designing a production machine by making both parts and labor interchangeable led to a mass-production system and a whole new competitive game. He could produce 6,000 cars a day, while his closest competitor in France could muster only 700 cars a year. The ability to produce increased by more than an order of magnitude. In one generation we produced goods and services that surpassed the cumulative capacity of mankind.

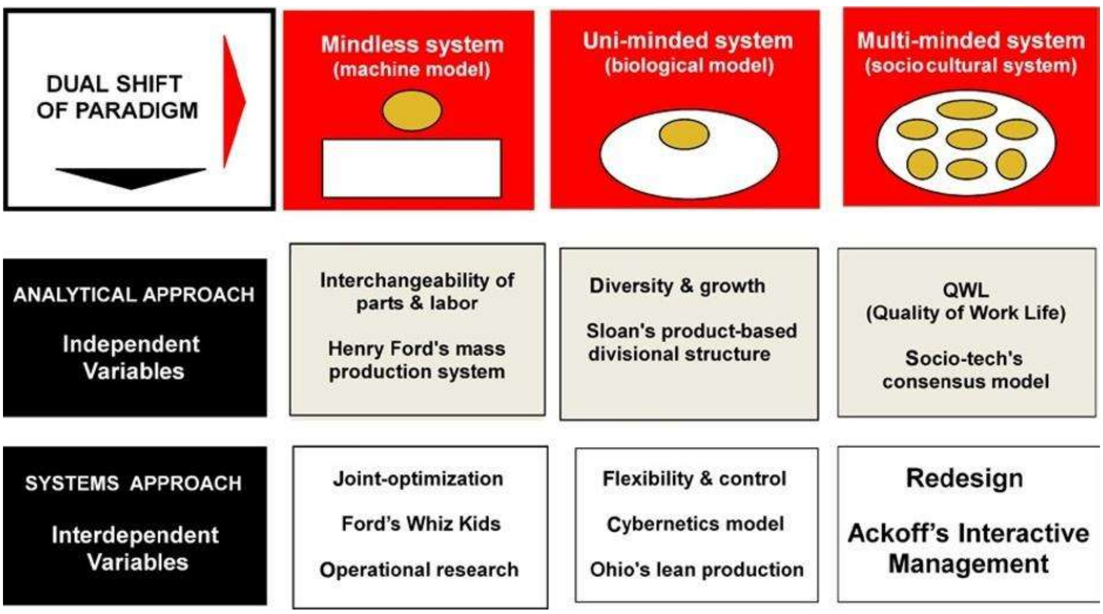

FIGURE 1.9 Six competitive games.

The effectiveness of this mode of organization in the production of goods and services created not just a quantitative change but also a qualitative change in the nature of the problem itself. The question was no longer how to produce, but how to sell. And so dawned the marketing era. What emerged was an environment with an entirely new set of challenges. Foremost among them was how to respond to increasing demand for variety and diversity, and how to manage growth in size and complexity.

This challenge was too great for even the best that a machine mode of organization could offer. The requirement for no deviation, in view of the assumption that human nature is essentially deviant, places high emphasis on tight supervision to ensure conformity, predictability, and reliability of individual behavior within the organization. This emphasis undermines the organization's creative ability and limits its response to meeting the increasing demand for variety and diversity. A defensive reaction to consumer dissatisfaction calls for greater adherence to the rules and more rigidity, resulting in a vicious circle.

On the other hand, growth in size tends to reduce efficiency and organizational effectiveness. Because of an inverse relationship between an organization's size and the effectiveness of its control system, large organizations are forced toward decentralization. But this result is inconsistent with the principle of no deviation and unity of command.

No driver in his or her right mind would drive a car with decentralized front wheels. In an organization that demands a passive functioning of parts with a high degree of compatibility and predictability, decentralization leads to chaos and suboptimization. The best answer for production may be in conflict with the best answer for marketing, and may not necessarily agree with the best answer for finance or personnel. Could this be why most large organizations constantly oscillate between centralization and decentralization?

1.8.2 Divisional Structure — Managing Growth and Diversity

Unlike Ford, Sloan recognized that the basis for competition had changed from an ability to produce to an ability to manage growth and diversity. He not only used public financing to generate the necessary capital to sustain growth, but also capitalized on the emerging biological model to provide a structural vehicle for control that made it possible to manage growth and diversity.

Sloan's model, with small variations, constitutes the foundation of the MBA programs taught in all prominent schools of management, including Harvard, Wharton, Stanford, and MIT. Operationally, this model is built around two concepts: *divisional structure* and *predict-and-prepare mode of planning* (Figure 1.10).

Corporations, in their simplest form, are divided into two distinct types: corporate office and operating unit. A corporate office with a traditional functional structure is the “brain of the firm,” with an algorithm, which is

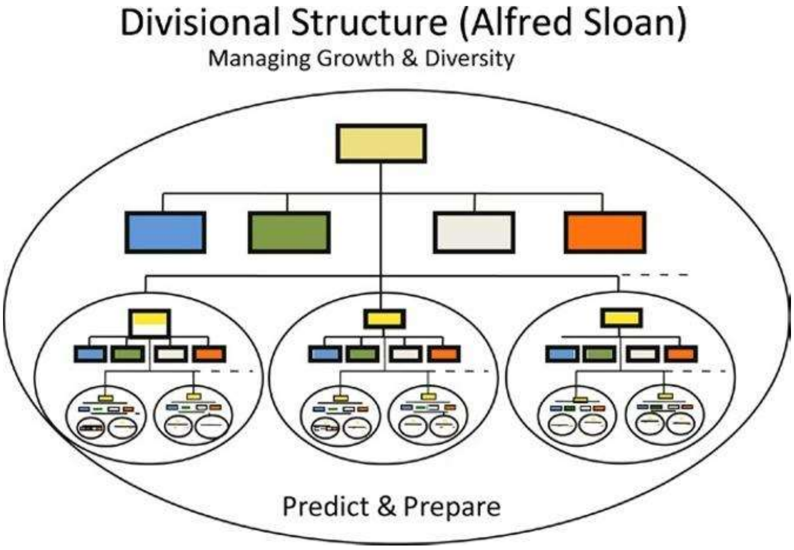

FIGURE 1.10 The divisional structure.

a procedure for producing a desired outcome and for monitoring its implementation. The operating unit, on the other hand, is the body, which, despite a semi-autonomous structure, has no choice and no consciousness. It can only react to the command signal from the brain and/or events in its environment. Ideally, an operating unit is a robot programmed to carry out, with no deviation, a set of procedures predefined by the functional units of the corporate office.

Replicas of this operating model — each a product division — are created as needed to produce a given product and/or service and sell it in a specified market. Operating product divisions are usually not authorized to redesign their products or redefine their markets. The main responsibility of the groups is to "stay the course." However, they are required to forecast the demand for their product and adjust their capacity to produce it accordingly. Therefore, the core concept of "predict and prepare" dominates the management process and complements the divisional structure in the pursuit of the essential functions: growth and viability.

The post-World War II environment, with its stability and predictability, provided an ideal condition for product-based divisional organizations. However, their very success in playing the game once again changed the game.

The divisional mode of organization, despite its unquestionable successes, found itself up against two unprecedented challenges:

1. The operationalization of new knowledge, in response to an overall shortening of product life cycles.
2. The reality of multi-mindedness, or understanding the implication of choice, and thus conflict, among the organization's members.

As a result of the research and development era, knowledge was generated at a faster rate, which called for periodically redesigning the product and redefining the markets. This capability, however, was incompatible with the mode of organization artfully designed to prevent change and stay the course. Successful divisional structure had tied the fate of product divisions to the life cycle of a single, predefined product. The division, then, like the product, experienced periods of uncertainty, growth, maturity, and decline. A popular solution for this concern, called *strategic planning*, dominated the practice of management in the United States for more than a decade. It simply called for identifying and assigning product divisions such designations as "question mark," "star," "cash cow," or "dog" and issuing imperatives to "drop the dog," "milk the cow," "watch the question mark," and "invest in the star." By default, it created the strategy of giving up on difficult challenges by simply tagging them dogs.

The divisional structure, finally, was challenged from two different directions: participative management and the lean production system. Both were emerging in tandem as alternative bases for new competitive games.

1.8.3 Participative Management

The unprecedented generation and distribution of wealth and knowledge resulted in ever higher levels of choice, which changed the nature of social settings and individual behavior in America. But the enhancement of choice, which resulted in higher levels of sophistication in social interactions, proved a double jeopardy for the biological mode of thinking. Not only did organizations conceived as uni-minded systems become more difficult to manage, but they also became more vulnerable to the actions of a few. Members of an organization, unlike the parts of a biological being, do not react passively to the information they receive.

In this regard, advances in information technology and communication as a means of control did not produce the panacea once expected. Even the ultimate in this mode of thinking, Stafford Beer's famous *Brain of the Firm* (1967), despite its elegance, in my experience, is unable to deal with the complexities of emerging social interactions. Nevertheless, the model was successful in the context of paternalistic cultures, where loyalty, conformity, and commitment are considered core virtues. These virtues are reinforced by the security of belonging to a group, which in turn protects and provides for its members. For example, Japan, an industrialized society, with a relatively strong paternalistic culture, closely approximates a uni-minded system. Therefore, it has been able to capitalize more effectively on the strength of the biological mode of organization.

In a strong paternalistic culture, conflict can be resolved by the intervention of a strong father figure, but the realities of highly developed multi-minded social systems are fundamentally different. Members of societies that have outgrown the secure, unifying web of a paternalistic culture display real choice. But a price must be paid for this transformation, especially in terms of insecurity and the level of conflict. The purposeful actors, individually or in groups, generate unprecedented levels of conflict by disagreeing with each other on the compatibility of their chosen ends and means.

Corporate America, yet ill-equipped to deal effectively with the consequences of its members' purposeful behavior, is finding itself increasingly paralyzed. It is not surprising that a significant part of its energy is lost to the conflict. Frustration associated with excessive levels of conflict reinforces the organizational inability to change. Members increasingly behave independently, and management, on the pretext of empowerment, abdicates its authority and responsibility. Nobody seems to have a handle on integration. Feelings of impotency and alienation are commonplace.

Pursuing the ideal of a conflict-free organization has proved problematic. Creating a conflict-free organization means less choice, reducing members to the level of robots. Such a situation, even if feasible, may not be desirable.

Unable to uncook eggs already half-cooked, we have rejected the paternalistic culture, but have not yet found an effective replacement for it. Unfortunately, quality of work life (QWL), participative management, multifunctional teams, and the other concepts that socio-tech had to offer have yet to show us how to manage a multi-minded complexity and effectively dissolve conflict. We are still oscillating between centralization and decentralization, collectivity and individuality, and integration and differentiation, without appreciating the complementary nature of these tendencies. We will deal with these issues in more detail in Part Two of this book.

The next three games represent the other dimension of the dual paradigm shift, dealing with the challenge of interdependency. They actually map the evolution of systems thinking in the context of mechanical, biological, and sociocultural models of organization.

1.8.4 Operations Research — Joint Optimization

The success of the first Operations Research (OR) group, created by Ackoff and Churchman at the Case Institute of Technology, which dealt with the challenge of interdependency, resulted in the spread of OR programs to most American universities. But the first full application of OR in corporate America came with Ford's whiz kids, when McNamara and his associates moved from the Defense Department to the Ford Corporation.

The essence of this effort was to use models, basically mathematical, to find optimal solutions to a series of interdependent variables. However, the assumptions regarding the nature of the organization remained mechanical. The other significant contribution to this version of systems thinking was the concept of systems dynamics developed by J. Forrester of MIT.

Operations Research dominated the field of systems thinking for the better part of the 1960s until it was challenged, ironically, by one of its founding fathers. In a famous article, Ackoff (1979) declared, “The future of Operations Research is past.” Instantaneously, he converted an army of devoted followers into staunch enemies. He blasted his own creation on the grounds that OR assumes passive or reactive parts and does not appreciate the vital implications of parts having choice.

By the assertion that parts in a social system have a choice, he left his contemporaries behind by a quarter of a century. His concept of multi-minded purposeful systems effectively bypassed the next generation of the systems models, most importantly Beer's *viable systems*, which in its own right is a masterful thinking in the biological context.

1.8.5 Lean Production System — Flexibility and Control

Effective commercial use of organized research, which evolved during World War II, accelerated the role of product development, giving rise to a new era marked by rapid change. Unpredictability associated with the high rate of change undermined the usefulness of the core concept of *predict* and *prepare*. Both the Chase and Wharton Econometric models, which had brought fame and fortune to their respective organizations, even a Nobel Prize for the Wharton School, were sold quietly.

The research and development era had generated explosions of new knowledge. This knowledge, when successfully operationalized, radically changed the competitive game. The new generation of winners were those players with the ability to create their own future by interactively influencing their environment. The name of the game became flexibility and control, which shortened the time to market of a new product, increased product/market differentiation, and improved price/quality performance of the outputs, doing more and more with less and less.

This game emerged slowly but effectively in Japan, when Ohno, Chief Engineer of Toyota, created the lean production system by applying systems thinking in the biological context. Using cybernetic principles, he was able to lower the break-even point by an order of magnitude and elevated the competitive game to an incredibly higher level. In this game, flexibility and control became the basis for competition.

1.8.6 Interactive Management — Design Approach

Design is the operational manifestation of the purposeful systems paradigm developed by Ackoff (1972) in response to the challenge of managing interactions between purposeful members of a highly interdependent social organization.

Systems design, at present, represents the latest chapter of the evolution of systems thinking. In *Redesigning the Future,* Ackoff (1974) argued that purposeful social systems are capable of recreating their future; they do so by redesigning themselves. Ackoff then proposed a design methodology by which stakeholders of a multi-minded system participatively design a future they collectively desire and realize it through successive approximation.

In *The Design of Inquiring Systems,* Churchman (1971) demonstrated that the best way to learn a system is to design it. Later, in *A Prologue to National Development Planning,* Gharajedaghi and Ackoff (1986) used design as the main vehicle of social development. The design model explicitly recognized that choice is at the heart of human development. Development is the enhancement of the capacity to choose; design is a vehicle for enhancement of choice and holistic thinking.

Designers seek to choose rather than predict the future. They try to understand rational, emotional, and cultural dimensions of choice and to produce a design that satisfies a multitude of functions. The design methodology requires that designers learn how to use what they already know, learn how to realize what they do not know, and learn how to learn what they need to know. Finally, producing a design requires an awareness of how activities of one part of a system affect and are affected by other parts. This awareness requires understanding the nature of interactions among the parts.

Unfortunately, despite all the rhetoric to the contrary, our risk models developed on the assumption of independency have failed to protect us against recurring events that have long been considered highly improbable. Nassim Taleb (2007), in his eye-opening book *The Black Swan,* demonstrated how in hindsight one could find a reasonable explanation for all of the following catastrophic events by appreciating interactions and powerful reinforcing effects of small interdependent deviations.

- 1982 recession (large American banks lost close to all their cumulative earnings)
- Real state collapse of early 1990s (savings and loans were wiped out at the cost of $500 billion).
- 1998 collapse of stock market (dotcom bubble)
- 2009 financial crisis (housing bubble and mortgage fiasco, possibly trillions of dollars)

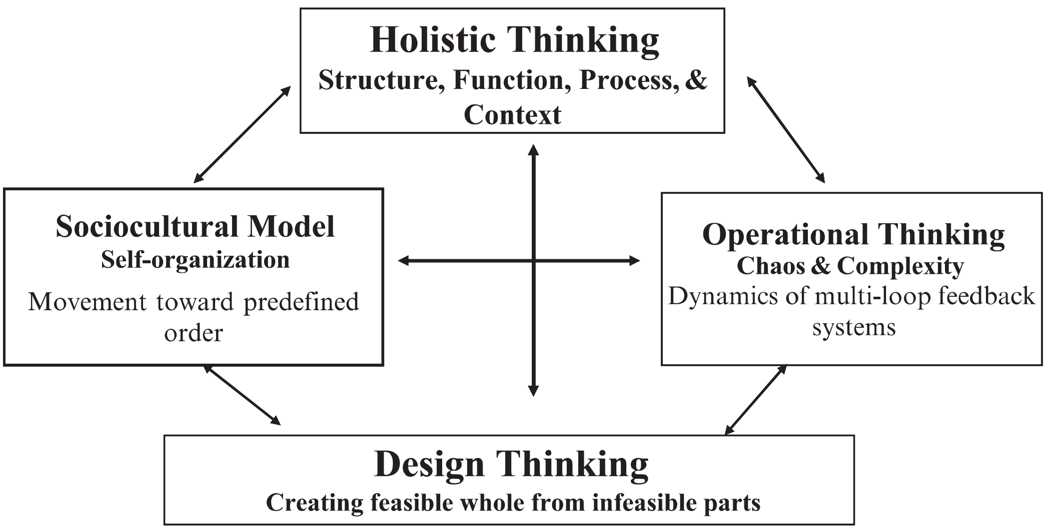

FIGURE 1.11 Foundations of systems thinking.

Unfortunately the task is not just an academic discourse; it demands enormous emotional struggles and a huge cultural challenge. Engagement in this process, in addition to competence, requires courage.

The remainder of this book attempts to explore the operational meaning of systems thinking and demonstrate the interaction of the four foundations of systems thinking seen in Figure 1.11. The task is also to create a comprehensive methodology that can meet the challenges of the emerging chaotic and complex environment.